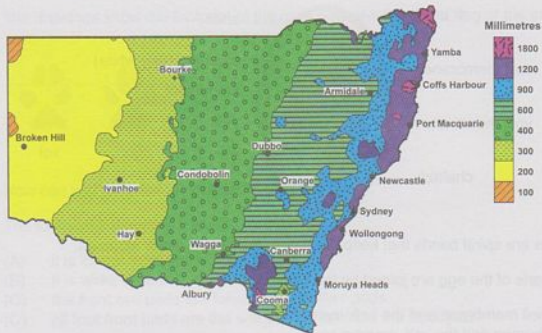


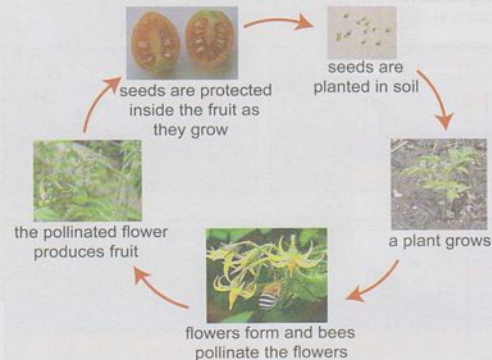
19. This map shows the annual rainfall across New South Wales.



Which of these towns receives approximately 900 millimetres of rainfall annually?

- (A) Dubbo (B) Orange
(C) Wollongong (D) Coffs Harbour

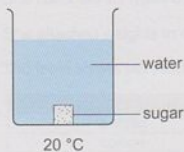
20. The photographs show the life cycle of a tomato plant.



The fruit of a tomato plant grows

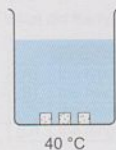
- (A) from the flower. (B) from the leaves.
(C) inside the seed. (D) under the soil.

21. Linda wanted to know if sugar dissolves faster in warm water than in cold water. This is how she set up the first part of her experiment.

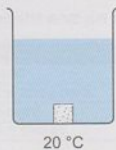


Which set-up should Linda use to correctly complete her experiment?

(A)



(B)



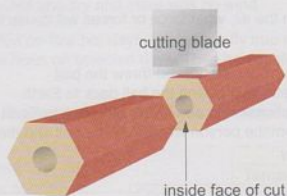
(C)



(D)



22. A pencil has been cut in half. This diagram shows what it looks like inside the cut.



Another pencil was cut in half as shown in the diagram.



What will the inside face of the cut look like?

(A)



(B)



(C)

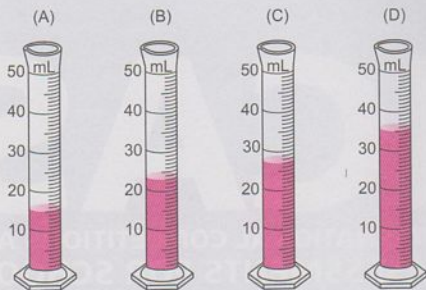


(D)



1. The picture shows some containers with coloured water in them.

Which container has 35 millilitres (mL) of water in it?



2. Jim had the idea that there were more flies than bees in his school grounds.

Here is a list of all the insects he saw over five days at school.

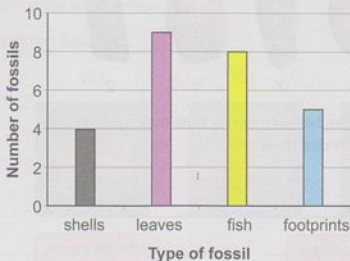
Day	Number of insects
Monday	• 5 cockroaches • 2 flies
Tuesday	• 2 cockroaches • 5 flies
Wednesday	• 1 stick insect • 6 flies
Thursday	• 2 bees • 2 cockroaches
Friday	• 2 beetles • 3 cockroaches

What should Jim do to find out whether his idea was right or wrong?

- (A) Compare the number of flies seen on Monday with the number of bees seen on Thursday.
- (B) Compare the number of flies seen on Tuesday with the number of bees seen on Thursday.
- (C) Compare the number of flies seen on Monday with the number of flies seen on Tuesday.
- (D) Compare the total number of flies seen over the five days with the total number of bees.

3. A geologist found different types of fossils while investigating a dry river bed.

The graph shows the number of each type of fossil she found.



How many fish fossils did she find?

- (A) 4 (B) 5 (C) 8 (D) 9

4. The table shows the vitamin C content of different fruits.

Fruit	Vitamin C (mg per 100 g)
apple	6
banana	9
mango	28
pear	4

Which of these fruits is the greatest source of vitamin C?

- (A) apple (B) banana (C) mango (D) pear

5. When seasons change we can see changes in the trees. In spring and summer, trees have a lot of green leaves. In autumn, the leaves often change colour. In winter many trees lose all of their leaves.

Which of these photos was most likely taken in winter?

(A)



(B)



(C)



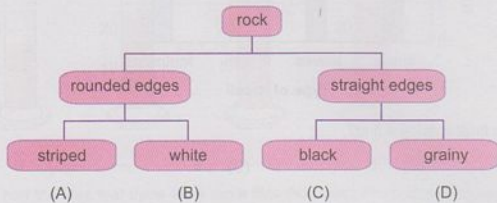
(D)



6. Wendy photographed four rocks W, X, Y and Z.



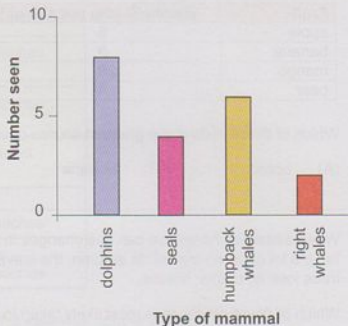
She used the differences between the rocks to make a chart.
Which path on the chart describes rock Y?



7. The graph shows the number of mammals seen on a whale watching tour.

How many seals were seen?

- (A) 2
(B) 4
(C) 5
(D) 6



8. The planet Saturn is surrounded by rings.
The planet Venus is surrounded by cloud.
The planet Mercury is covered in craters.
The planet Jupiter has no solid surface.

Which of these could be Saturn?

(A)



(B)



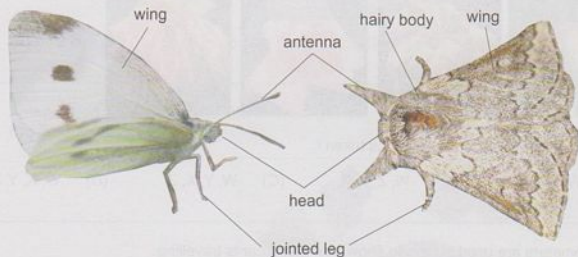
(C)



(D)



9. The photographs show a butterfly and a moth at rest. The butterfly is viewed from the side. The moth is viewed from above. Some body parts are labelled.



Which feature is common to both animals?

- (A) hairy body
(B) long thin antennae
(C) legs that are jointed
(D) position of wings when resting

10. Sean recorded information about some containers.

Container	What is kept in the container?	What is the container made of?
bottle	soft drink	plastic
jar	coffee	glass
carton	milk	cardboard
can	soup	metal
bottle	sauce	glass
tub	butter	plastic

What is kept in glass containers?

- (A) coffee and sauce (B) coffee and milk
(C) soup and sauce (D) soup and milk

11. The photographs show the same flower at different stages of its life.



In which order were the photographs taken?

- (A) W, Y, Z, X (B) W, Z, Y, X (C) W, Y, X, Z (D) W, X, Y, Z

12. Speedometers are used in cars to show how fast a car is travelling.



At what speed is this car travelling?

- (A) 20 km/h (B) 25 km/h (C) 30 km/h (D) 35 km/h

13. What is the height of this stack of wooden blocks?



- (A) 2 cm (B) 5 cm (C) 7 cm (D) 10 cm

14. Abdul sorted some sea shells into different groups. The photographs show the two groups.



Which names could describe the shells in these groups?

- (A) long shells and short shells
(B) rounded shells and pointed shells
(C) wing-shaped shells and snail-shaped shells
(D) dark-coloured shells and light-coloured shells

15. People sometimes confuse fox and dog footprints.

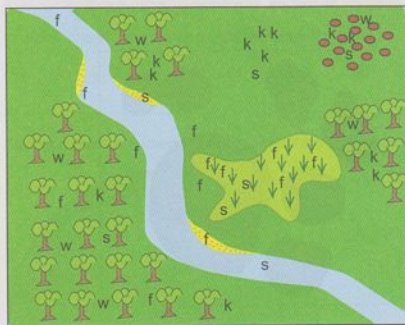
The drawings show the footprint of the front paw of a fox and a dog of the same size.



How can you tell a fox footprint from that of a dog?

The footprint belongs to a fox if

- (A) it is longer than that of the dog.
 (B) it is wider than that of the dog.
 (C) the front two pads are forward of all other pads.
 (D) all four front pads are the same distance from the back pad.
16. Jai drew a map of his local area to show where
- particular environments exist
 - individual animals have been observed.

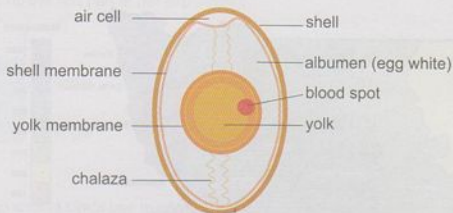


KEY	
Environments	
	grasslands
	swamp lands
	rocky area
	river
	wood lands
	sandy beach
Animals	
f	frog
k	kangaroo
s	snake
w	wombat

Which animal has been observed in all six environments in Jai's local area?

- (A) frog (B) snake (C) wombat (D) kangaroo

17. The drawing shows the parts of a hen's egg.

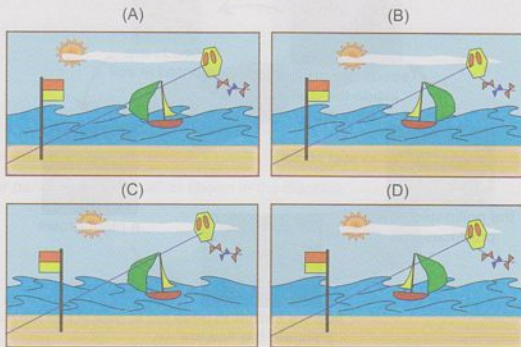


The chalazas are spiral bands that keep the yolk in the centre of the albumen.

Which two parts of the egg are joined by the chalazas?

- (A) the shell membrane and the yolk membrane
 (B) the albumen and the yolk membrane
 (C) the albumen and the yolk
 (D) the shell and the yolk
18. The wind will cause the sea and the boats on it, as well as kites and flags on the beach, to move in the same direction.

Which drawing correctly shows all movement caused by the wind?



23. The Sun is a yellow, medium-sized star.

The temperature of a star can be worked out by its colour.

Here are some examples:

Colour	Temperature (°C)
blue	50 000 – 28 000
blue-white	28 000 – 10 000
white	10 000 – 7 500
white-yellow	7 500 – 6 000
yellow	6 000 – 4 900
orange	4 900 – 3 500
red	3 500 – 2 000

Based on this information, what is the temperature of the Sun?

- (A) 28 000 – 10 000 °C
- (B) 10 000 – 7 500 °C
- (C) 7 500 – 6 000 °C
- (D) 6 000 – 4 900 °C

24. A force can be a push or a pull.

The speed and direction of an object can only be changed by a force.

A basketball is thrown in the air on a day when there is no wind.

Once the basketball is in the air, what force or forces will cause the basketball to change direction?

- (A) the initial push from the person who threw the ball
- (B) the Earth's magnetic field pulling the ball back to Earth
- (C) the Earth's gravitational field pulling the ball back to Earth
- (D) the initial push from the person who threw the ball and then gravity pulling the ball back to Earth

25. Year 3 made some model cars. They wanted to test the cars to find out which one went the fastest. Each student let a car go from the top of a ramp and timed how long it took for the car to reach the bottom.

For the test to be fair, the students need to make sure that

- (A) each student uses a different stopwatch.
- (B) the height of the ramp stays the same.
- (C) each student repeats the experiment.
- (D) the cars are all the same colour.

26. Kerry tested the strength of different threads.

She collected six types of thread and fastened them to the back of a chair.

She attached weights to each piece of thread until the thread broke.

The table shows her results.

Type of thread	Breaking weight (g)
acrylic	250
cotton	25
linen	500
nylon	50
silk	200
wool	10

What did Kerry change and what did she measure in her experiment?

	Kerry changed	Kerry measured
(A)	the type of thread	the breaking weight
(B)	the height of the chair back	the length of the thread
(C)	the breaking weight	the type of thread
(D)	the length of the thread	the height of the chair back

27. Vickram thought that flies were present outside his house in greatest numbers when the weather conditions were hot and dry and there was no wind.

He wanted to test this idea on four hot days. He hung a fly trap under the same tree in his garden at 11 am, for one hour, on selected days. Then he counted the number of flies in the trap.

To make this a fair test, Vickram knew he would need to repeat this process in different hot conditions.

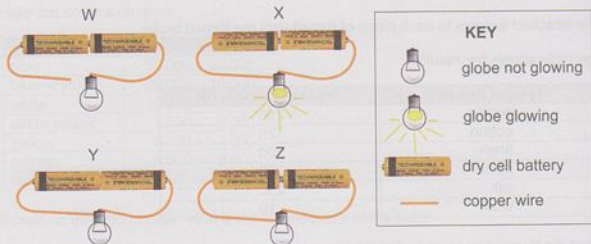
Vickram listed the different combinations of weather conditions he would need to test.

	Weather conditions		
	Temperature	Rain	Wind
Condition 1	hot	dry	calm
Condition 2	hot	wet	calm
Condition 3	hot	dry	windy

What further conditions does Vickram need to investigate to achieve a fair test?

- (A) one hot, wet and windy day
- (B) three cold days, all wet and two windy
- (C) no further conditions need to be tested
- (D) repeat the experiment using the same conditions

28. Vanessa was investigating how she could make an electrical circuit using two batteries, some wire and a globe. She used four different arrangements, W, X, Y and Z.



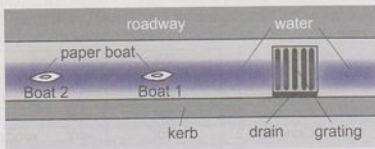
Only arrangement X made the globe light up.

She concluded that the globe will glow only when the batteries are facing in

- the same direction in the circuit and the circuit is open.
- the same direction in the circuit and the circuit is closed.
- opposite directions in the circuit and the circuit is open.
- opposite directions in the circuit and the circuit is closed.

29. Luka floated two identical paper boats on water in a gutter. The water was flowing towards a drain.

The drawing shows the position of the boats at a particular moment in time.



Boat 1 went down the drain 10 seconds after this.

What most likely happened to Boat 2?

- It remained in the same position.
- It went down the drain about 5 seconds after Boat 1.
- It went down the drain about 10 seconds after Boat 1.
- It went down the drain about 20 seconds after Boat 1.

30. Objects either sink or float when placed in water.

An object will sink in water if the mass of the object is greater than the mass of water with identical volume to that of the object.

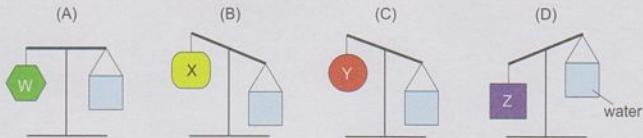
Objects W, X, Y and Z have identical volumes.

The diagram shows what happened when the objects were placed into a tank of water.



The diagrams below show what happened when the mass of each object was compared to the mass of an identical volume of water.

Which diagram is **not** correct?





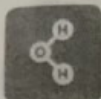
SC

2009 Science Answer Keys

ICAS
International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Paper I	Paper J
1	D	A	C	A	C	A	A	B	D	D
2	D	A	B	C	B	B	A	A	D	D
3	C	C	C	A	A	A	C	C	B	B
4	C	D	D	B	D	A	C	B	D	D
5	D	B	B	B	A	D	C	D	C	C
6	A	C	D	B	B	A	C	B	C	C
7	B	A	B	C	D	C	A	C	B	B
8	D	B	A	D	B	B	C	B	B	B
9	C	A	A	B	A	D	A	B	B	B
10	A	D	A	A	B	B	B	B	B	B
11	A	C	D	D	D	A	B	C	C	C
12	C	B	C	A	A	A	D	A	D	D
13	D	B	D	C	B	C	D	C	A	A
14	C	D	C	C	A	C	B	D	D	D
15	C	A	B	D	A	B	B	B	C	C
16	B	C	A	B	C	C	B	D	D	D
17	A	A	A	D	C	D	D	A	C	C
18	B	D	B	D	A	D	B	A	C	C
19	B	D	D	B	C	D	A	A	A	A
20	A	B	B	A	D	C	A	B	A	A
21	D	D	A	D	D	B	C	B	C	C
22	D	B	B	B	C	C	B	C	A	A
23	D	A	C	C	C	D	C	C	A	A
24	C	C	A	D	C	C	C	A	B	B
25	B	C	C	C	C	B	A	C	A	A
26	A	C	B	B	C	A	B	A	A	A
27	A	D	D	D	D	C	C	D	D	D
28	B	D	C	C	A	C	D	C	C	C
29	C	B	D	C	D	B	B	A	B	B
30	B	A	D	B	B	B	D	B	A	A

(Please turn over)



SC | 2009 Science Answer Keys



Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Paper I	Paper J
31			D	B	A	A	C	D	B	B
32			C	C	C	D	A	D	B	B
33			B	A	D	B	B	D	B	B
34			C	D	D	D	B	B	D	D
35			A	B	B	D	A	D	C	C
36			D	A	B	A	B	A	D	D
37			B	A	B	B	B	D	D	D
38			A	D	B	C	D	C	D	D
39			C	A	D	D	D	D	C	C
40			D	B	D	D	A	A	B	B
41					C	C	D	A	A	A
42					B	C	B	D	C	C
43					B	D	C	C	B	B
44					D	D	D	C	B	B
45					C	C	D	B	D	D

Contact Details

Tel: (02) 8344 1016
Email: info@eaa.unsw.edu.au
Web: www.eaa.unsw.edu.au